



Sunbeam Institute of Information Technology

Pune and Karad

Module – Internet of Things (IoT)

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- What is AI?
- Applications of AI in IoT
- Data Science
- Machine learning - types
- numpy, pandas
- Supervised - linear regression / SVM / KNN
- Deep learning - Neural Networks

Tensor Flow / PyTorch



Keras

- Feature Engg
- Ensemble learning
- Time series & forecasting

When did it start?

- Greek Mythology – Talos
 - Talos was a giant animated bronze warrior programmed to guard the island of Crete
- 1950 – Alan Turing
 - Alan Turing published a landmark paper in which he speculated about the possibility of creating machines that think
 - What he created is known as Turing Test which is used to determine whether or not the computer can think intelligently like human being
- 1951 – Game AI
 - Christopher Strachey wrote a checkers program and Dietrich Prinz wrote one for chess
- 1956 – The birth of AI
 - John McCarthy first coined the term Artificial Intelligence at Dartmouth Conference
- 1959 – First AI laboratory
 - MIT AI lab was first set up in 1959 and research on AI began

When did it start?

- 1960 – General Motors Robot
 - First robot was introduced to General Motors assembly line
- 1961 – First chatbot
 - The first AI chatbot called ELIZA was introduced in 1961
- 1997 - IBM Deep Blue
 - IBM's Deep Blue beats world champion Garry Kasparov in the game of chess
- 2005 - DARPA Grand Challenge
 - Stanford Racing Team's autonomous robotic car, Stanley wins the 2005 DARPA Grand Challenge
- 2011 – IBM Watson
 - IBM's question answering system, Watson, defeated the two grated Jeopardy champions Brad Ruther and Ken Jennings

What is Artificial Intelligence

- Artificial Intelligence (AI), sometimes called machine intelligence, is intelligence demonstrated by machines, in contrast to the natural intelligence displayed by humans
- Any device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals
- The theory and development of computer system able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision making and translation
- Often used to describe machines (or computers) that mimic "cognitive" functions that humans associate with the human mind, such as "learning" and "problem solving"

learning & problem solving

- Simulating higher functions of the human brain
- Programming a computer to use general languages ← NLP
- Arranging hypothetical neurons in a manner so that they can form concepts ← DL-NN
- A way to determine and measure problem complexity
- Self-improvement more data – more learning
- Abstraction: defined as the quality of dealing with ideas rather than events
- Randomness and creativity

└ Generative AI – LLM

- **Why are we talking about it now ?**
 - More Computational Power
 - More Data
 - Better algorithms
 - Broad investment

- Google's search engine
- JPMorgan Chase's Contract Intelligence (COiN) platform uses AI, machine learning and image recognition software to analyse legal documents ← text processing - NLP
- IBM Watson: Healthcare organizations use IBM AI (Watson) technology for medial diagnosis
- Google's AI Eye Doctor can examine retina scans and identify a condition called as diabetic retinopathy which can cause blindness ← image processing - Computer Vision
- Facebook uses ML and DL to detect facial features and tag your friends
- Twitter uses AI to identify hate speech and terroristic language in the tweets - NLP
- Smart Assistants: Siri, Google Assistant, Alexa, Cortana, Gemini
- Tesla automated cars ← reinforcement learning
- Netflix uses AI for movie recommendations ← recommendation Engine
- Spam filtering ← classification

- Autonomous Vehicles
- Smart Home Devices
- Healthcare Devices
- Industrial Automation
- Agricultural Technology
- Retail and Supply Chain Management
- Energy Management
- Consumer Electronics
- Aerospace and Defense
- Environmental Monitoring

- **Edge AI**

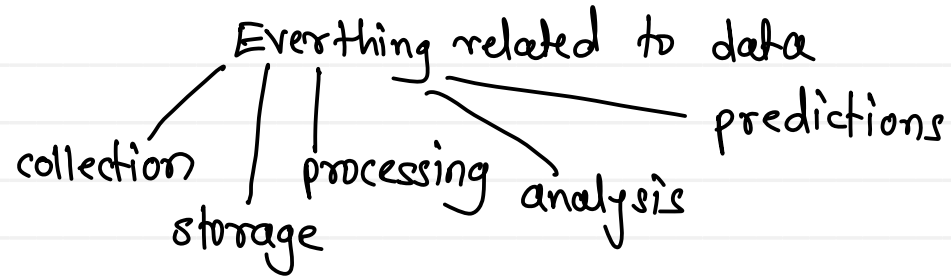
- Edge AI means doing computer stuff right on the device instead of sending it to the cloud. This makes things faster, keeps your data private, and uses less internet.

- **AIoT (AI and IoT)**

- When AI and the Internet of Things (IoT) come together, devices get smarter and work together better, making things easier and faster.

- **TinyML**

- Tiny Machine Learning (TinyML) helps put smart learning on devices with limited resources. So, this makes AI work on more things like gadgets and machines.



abc, 40, pune, 412101

↳ raw data → meaning less

↓ processing

name	age	city	pincode
abc	40	pune	412101

↳ information — meaningful

Data

1) structured data

— Database (MySQL)

2) Semi-structured data

— json/xml/csv

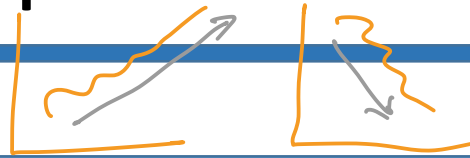
3) Unstructured data

— text, images, audio, video

What is Data Science?

- Data science is an inter-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from many structural and unstructured data.
- Data science is related to data mining, machine learning and big data.
- Data science is a "concept to unify statistics, data analysis and their related methods" in order to "understand and analyze actual phenomena" with data pattern.
- It uses techniques and theories drawn from many fields within the context of mathematics, statistics, computer science, domain knowledge and information science.

Quick overview of the process



Data Team

- Make the data available
- Clean the data

Business Intelligence Team

- Find business insights
- Prepares the dashboard

Data Science Team

- Use data analytics tools
- Develop models for various tasks

data collection :
├→ primary
└→ secondary

Analysis (past data)

- Analysis is performed on the things that are already happened in the past
- We do Analysis to explain How and or Why something happened
- E.g. Analyzing data by separating into chunks

Analytics (future data)

- Refers to the future after finding patterns
- We use Analytics to explore potential future events
- Branches
 - Qualitative
 - Intuition and experience forecasting
 - Analysis
 - Quantitative
 - Formulas
 - Algorithms

- Raw data (meaningless)
 - Can not be analyzed straight away
 - It is untouched data that is accumulated and stored on the server
 - Also known as raw facts or primary data
 - Can be collected by various techniques like
 - Survey
 - Automated tools

Data Pre-Processing

- This process tries to fix the problem that has occurred while data gathering
- Before processing with data analysis, it is important to remove the wrong data
- Techniques
 - Class Labeling
 - Labeling the data to the correct data types
 - e.g. numeric and categorical
 - Data cleansing
 - Deal with inconsistent data *missing data*
 - Also known as data cleaning or data scrubbing
 - Dealing with missing values
 - Data balancing
 - Data shuffling
 - Prevents unwanted patterns
 - Improves predictive performance

fourty → 40

- Once the data is cleaned and formatted, it can be analyzed for various reasons
- It explains past performance
- It can answer simple questions like
 - What happened ?
 - When did it happen?
- Or it can answer complex questions like
 - How did marketing team performed last quarter in terms of revenue
 - How does that compare to the performance in the same quarter last year

Predictive Analytics - Traditional (statistics)

- After the analysis is over, the next logical step is analytics ← future data
- It can be performed traditional statistical modelling like
 - Regression ← formula ← numeric value
 - A model used for quantifying causal relationships among the different variables included in your analysis ← independent
F dependent
 - Mostly used for predicting future values
 - Clustering
 - Creating different clusters (groups) by understanding data
- Factor Analysis
- Time Series analysis

known data

x	y
1	1
2	4
3	9
4	16
5	25
10	?

→ training data
→ testing data

① $x=1, y=1$

$y=x$

② $x=2, y=4$

$y=x$ $4 \neq 2$

$y=x+2$ $4 = 2+2$

③ $x=3, y=9$

$y=x+2$ $9 \neq 3+2$

$y=x^2$ $9 = 3^2$

④ $x=4, y=16$

$y=x^2$ $16 = 4^2$

$y=x^2$

→ formula
→ equation
→ model

Regression

Test the model

$x=5, y=25$

$y=x^2$ $25 = 5^2$

Prediction

$x=10$ $y=?$

$y=x^2$ $y=10^2$
 $=100$

REGRESSION

Exp	Salary
1.1	50 K
1.5	55 K
2.0	60 K
⋮	
10	1.5 L
12.5	2.

past data

known data

analyze

Patterns

model building or
finding formula

model

← relationship between
dependent & independent
variables

dependent variable (y):

- the value depends on input variables
- it will be always single variable

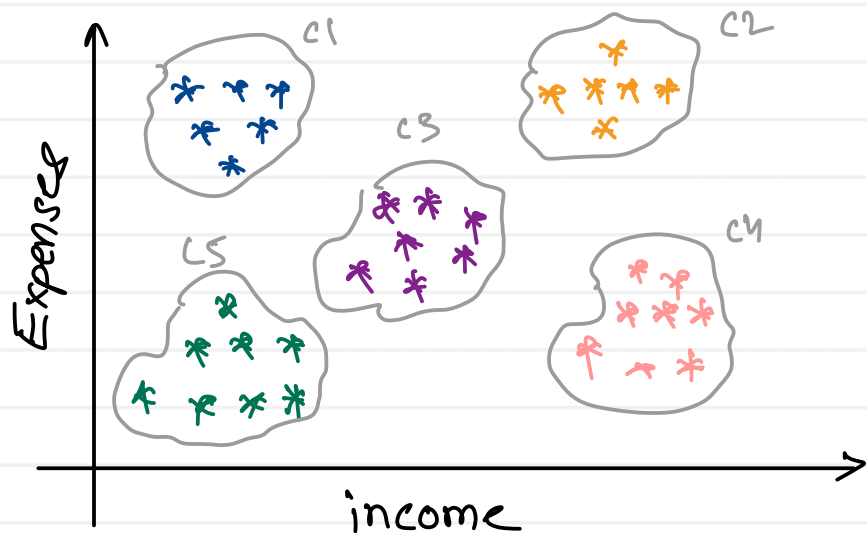
independent variable (x):

- these variables are used to find relationship or equation / formula
- single or multiple

change in experience causes change in salary
causal relation

- Exp & salary are
variables/attributes/
columns/features

CLUSTERING



cluster - group of values
clustering - process of
creating a group

c1 - low income, high expenses
c2 - high income, high expenses
c3 - med income, med expenses
c4 - high income, low expenses
c5 - low income, low expenses

Predictive Analytics – Machine Learning

- Utilizes artificial intelligence to predict behavior in unprecedented ways
- There are different techniques
 - Supervised Learning ← Regression, classification
 - Unsupervised Learning ← clustering, association
 - Reinforcement Learning

Artificial Intelligence:

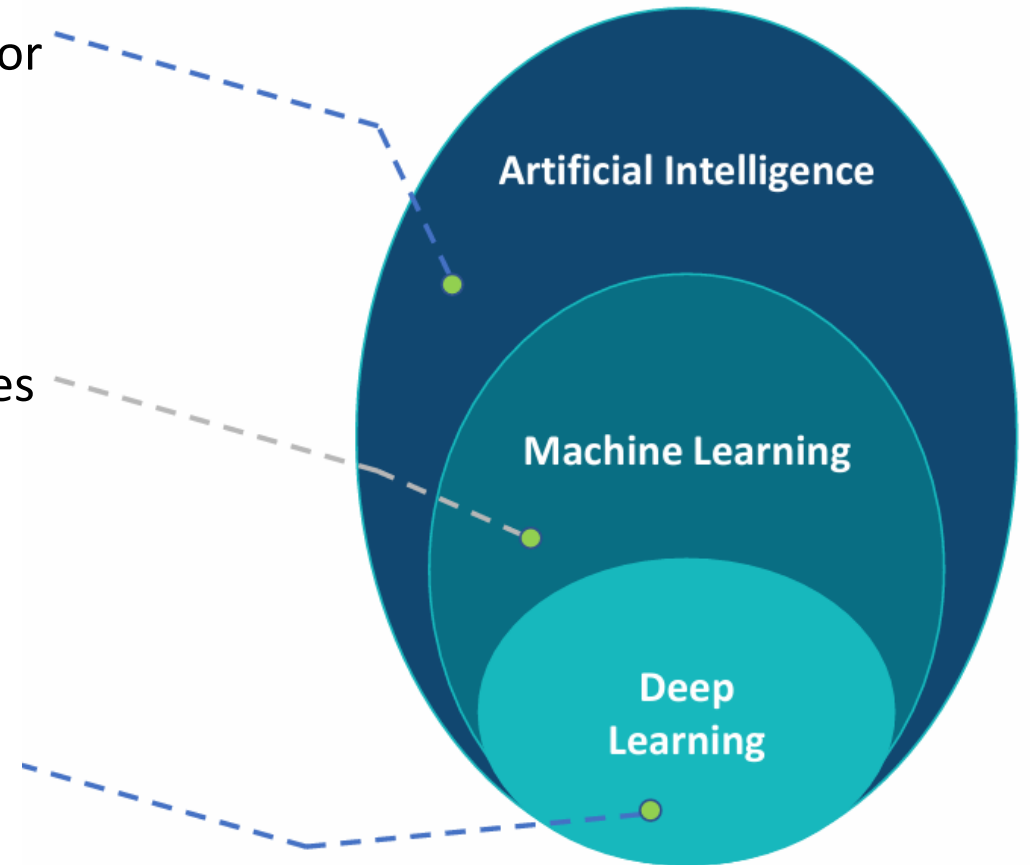
- A technique which enables machine to mimic human behavior

Machine Learning:

- Subset of AI which uses statistical methods to enable machines to improve the experience

Deep Learning:

- Subset of ML which makes the computation of multi-layer neural network feasible



What is machine learning ?

- A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E
 - Tom Mitchell, 1997
- Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed
 - Arthur Samuel, 1959
- Machine Learning is the science (and art) of programming computers so they can learn from data

Traditional approach — formula is known
— try to find answer for input

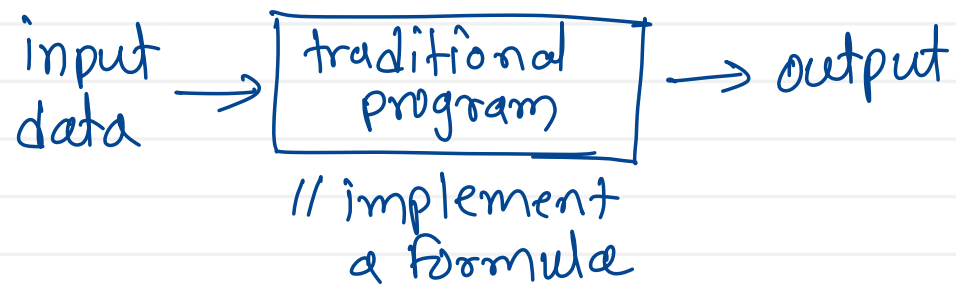
ML approach — formula is not known
↳ it is derived from past data
↳ also called as model
↳ model is used to find answers

Traditional approach

- ① take email as input
- ② check the source [IP Address]
- ③ check for spam keywords

```

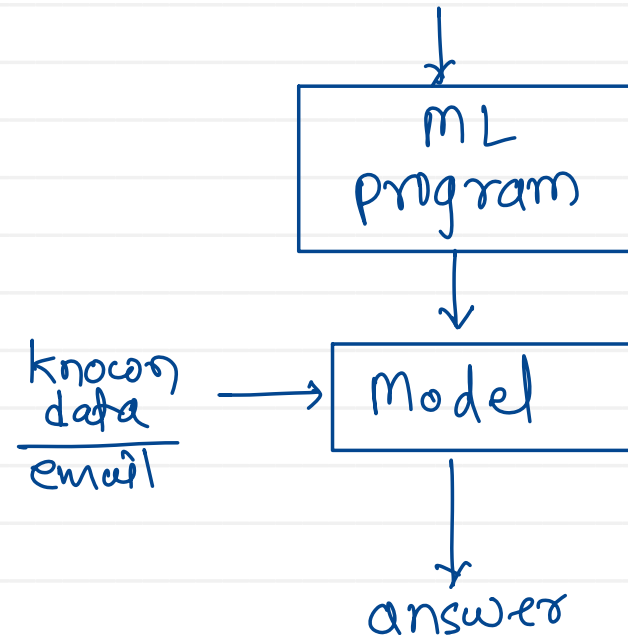
if( ... ) {
    // mark spam
} else if( ... ) {
    // mark spam
} else {
    // mark ham
}
    
```



Machine Learning approach

email	isSpam
... offer	1
-----	0

known data



- Analyzing images of products on a production line to automatically classify them
 - This is image classification, typically performed using convolutional neural networks
- Detecting tumors in brain scans
 - This is semantic segmentation, where each pixel in the image is classified (typically use CNNs)
- Automatically classifying news articles
 - This is natural language processing (NLP), and more specifically text classification
- Automatically flagging offensive comments on discussion forums
 - This is also text classification, using the same NLP tools
- Forecasting your company's revenue next year, based on many performance metrics
 - This is a regression task (i.e., predicting values) that may be tackled using any regression model
- Making your app react to voice commands
 - This is speech recognition, which requires processing audio samples: since they are long and complex sequences, they are typically processed using RNNs, CNNs, or Transformers

- Detecting credit card fraud
 - This is anomaly detection example
- Segmenting clients based on their purchases so that you can design a different marketing strategy for each segment
 - This is clustering example
- Representing a complex, high-dimensional dataset in a clear and insightful diagram
 - This is data visualization, often involving dimensionality reduction techniques
- Recommending a product that a client may be interested in, based on past purchases
 - This is a recommender system
- Building an intelligent bot for a game
 - This is often tackled using Reinforcement Learning

Where to use machine learning ?

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules:
 - one Machine Learning algorithm can often simplify code and perform better than the traditional approach
- Complex problems for which using a traditional approach yields no good solution:
 - the best Machine Learning techniques can perhaps find a solution
- Fluctuating environments:
 - a Machine Learning system can adapt to new data
- Getting insights about complex problems and large amounts of data

Types of machine learning

- There are so many different types of Machine Learning systems that it is useful to classify them in broad categories, based on the following criteria
 - Whether or not they are trained with human supervision
 - supervised, unsupervised, and Reinforcement Learning
 - Whether or not they can learn incrementally on the fly
 - online versus batch learning
 - Whether they work by simply comparing new data points to known data points, or instead by detecting patterns in the training data and building a predictive model, much like scientists do
 - instance-based versus model-based learning

Supervised Learning

- The majority of practical machine learning uses supervised learning
- Supervised learning is where you have **input variables (x)** and an **output variable (Y)** and you use an algorithm to learn the mapping function from the input to the output

$$Y = f(X) \rightarrow \text{model / formula}$$

- The goal is to approximate the mapping function so well that when you have new input data (x) that you can predict the output variables (Y) for that data
- It is called supervised learning because the process of an algorithm learning from the training dataset can be thought of as a teacher supervising the learning process
- We know the correct answers, the algorithm iteratively makes predictions on the training data and is corrected by the teacher
- Learning stops when the algorithm achieves an acceptable level of performance (measured in terms of accuracy)

- Regression *forecasting*
 - Related to predicting future values
 - E.g.
 - Population growth prediction
 - Expecting life expectancy
 - Market forecasting/prediction
 - Advertising Popularity prediction
 - Stock prediction
 - Algorithms
 - Linear and multi-linear regression
 - Logistic regression
 - Naïve Bayes
 - Support Vector Machine

- **Classification**

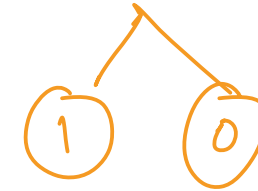
- Related to classify the records
- Based on class / labels (eg. Email : Spam / Ham , Gender : Male / Female , Loan : Yes / No)
- E.g.

- Find whether an email received is a spam or ham
- Identify customer segments
- Find if a bank loan is granted
- Identify if a kid will pass or fail in an examination

- **Algorithms**

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-nearest neighbor

class / label



- ① Binary classification
- ② multi class
↳ more two classes

- Unsupervised learning is where you only have input data (X) and no corresponding output variables
- The goal for unsupervised learning is to model the underlying structure or distribution in the data in order to learn more about the data
- These are called unsupervised learning because unlike supervised learning above there is no correct answers and there is no teacher
- Algorithms are left to their own devices to discover and present the interesting structure in the data
- Structure in the form of GROUPS / CLUSTERS / ASSOCIATION
- Mostly used for EDA (Exploratory Data Analysis)

- ① No model / formula is created (since we don't have output variable)
- ② No evaluation

- **Clustering**
 - discover the inherent groupings in the data, such as grouping customers by purchasing behaviour
 - E.g.
 - Batsman vs bowler
 - Customer spending more money vs less money
 - Algorithms
 - K-means clustering
 - Hierarchical clustering
 - DB Scan

- **Association**

- An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y
- E.g.
 - Market basket analysis
- Algorithms
 - Apriori
 - Eclat



Thank you!!!

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